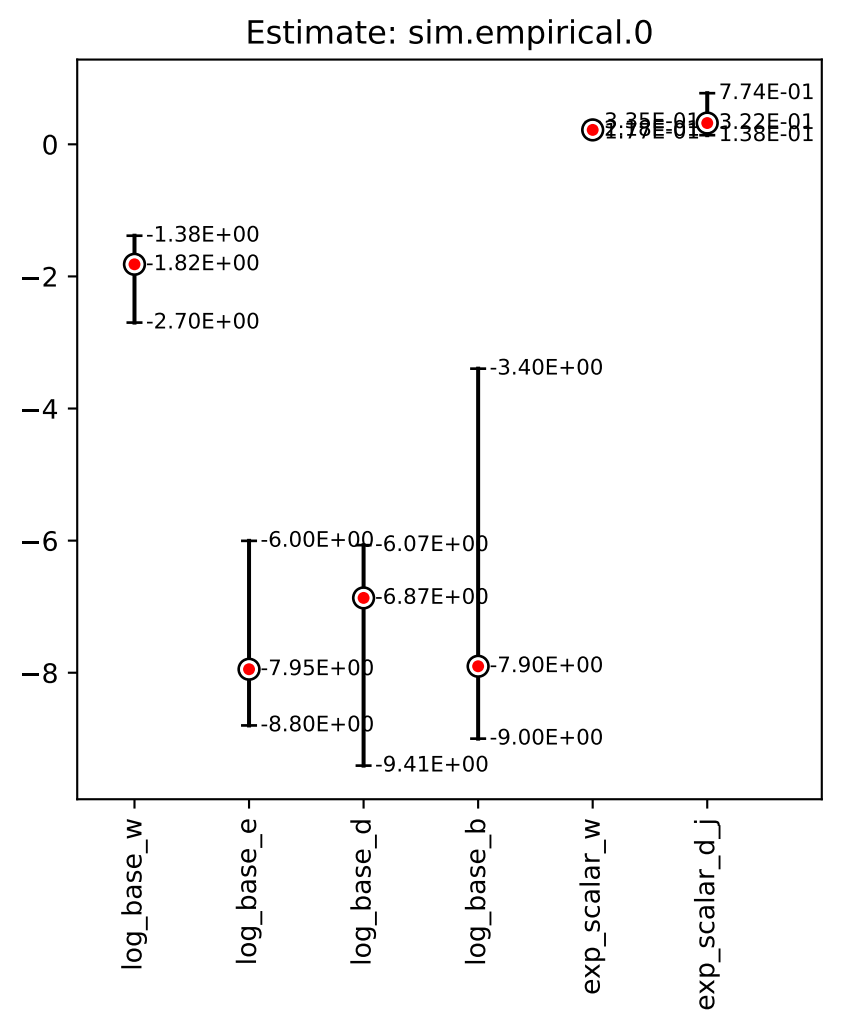
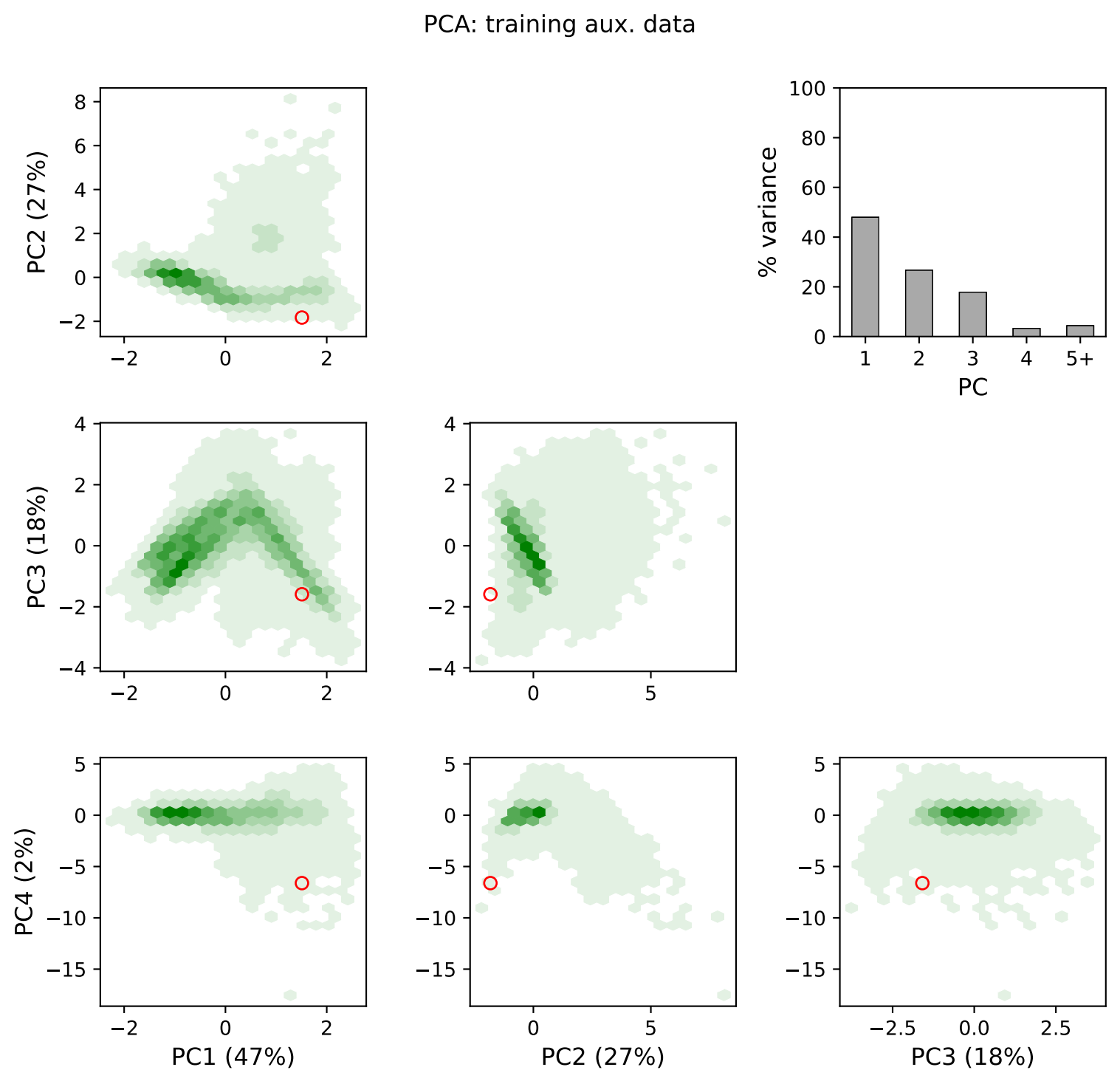
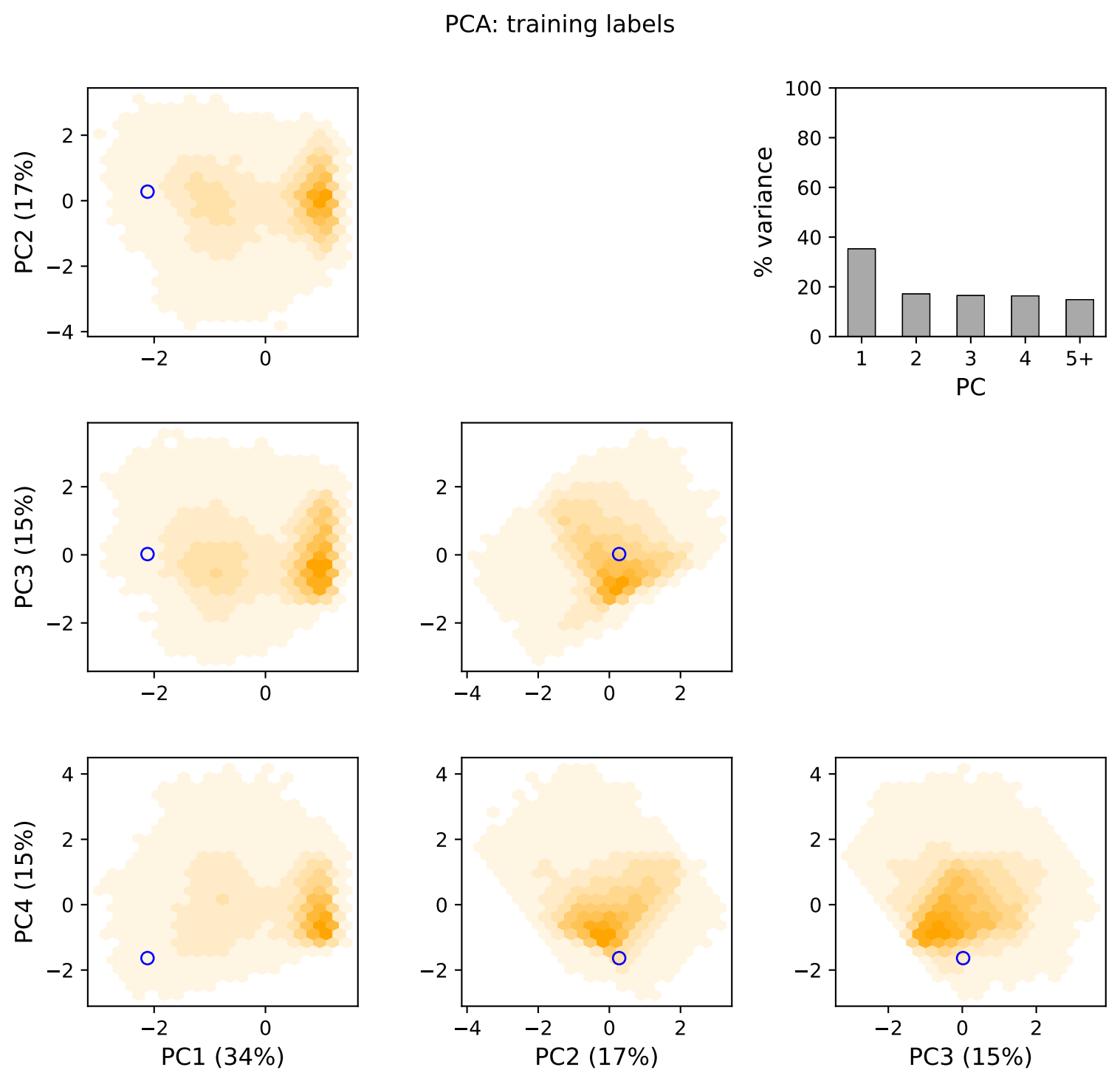
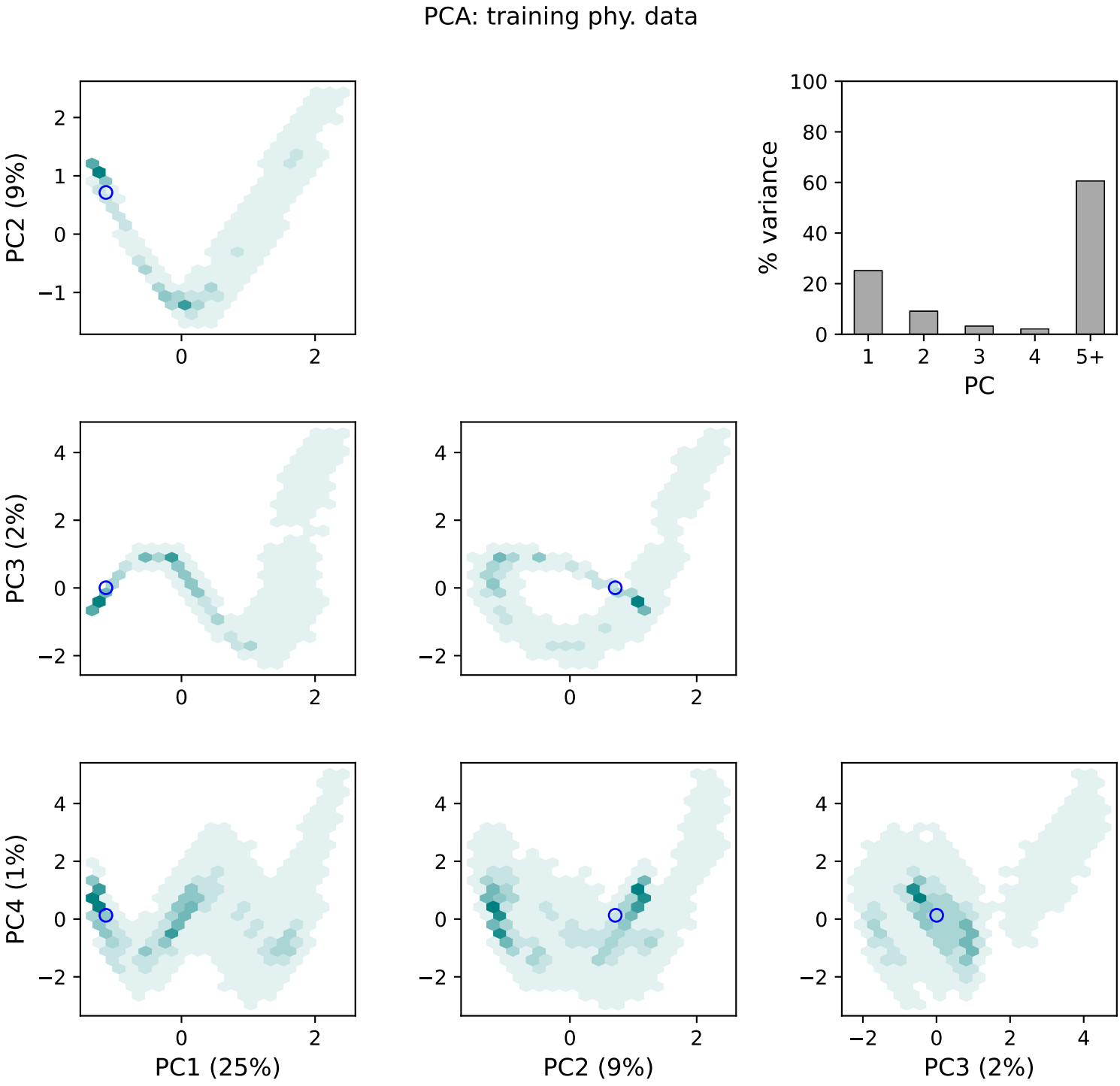
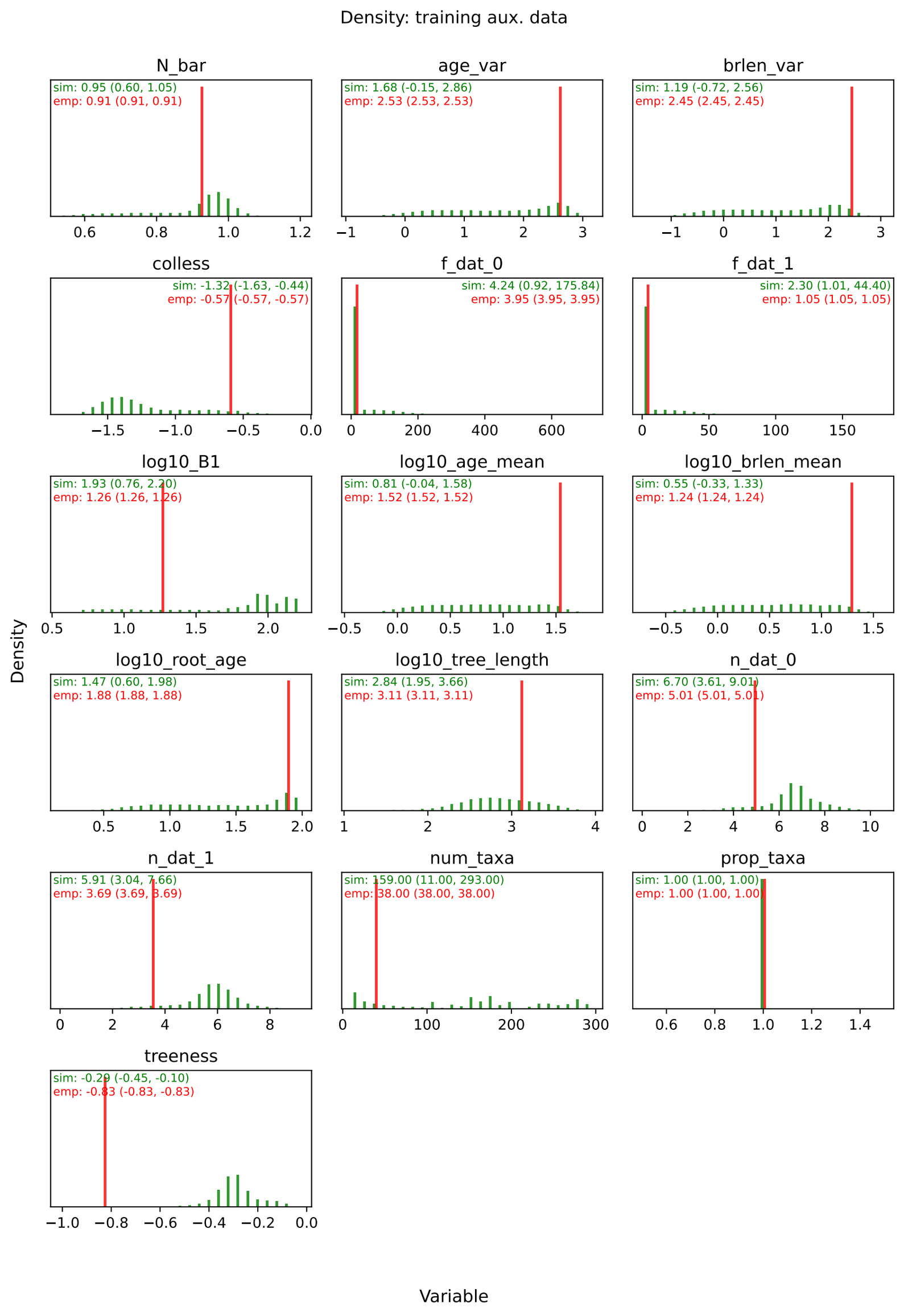
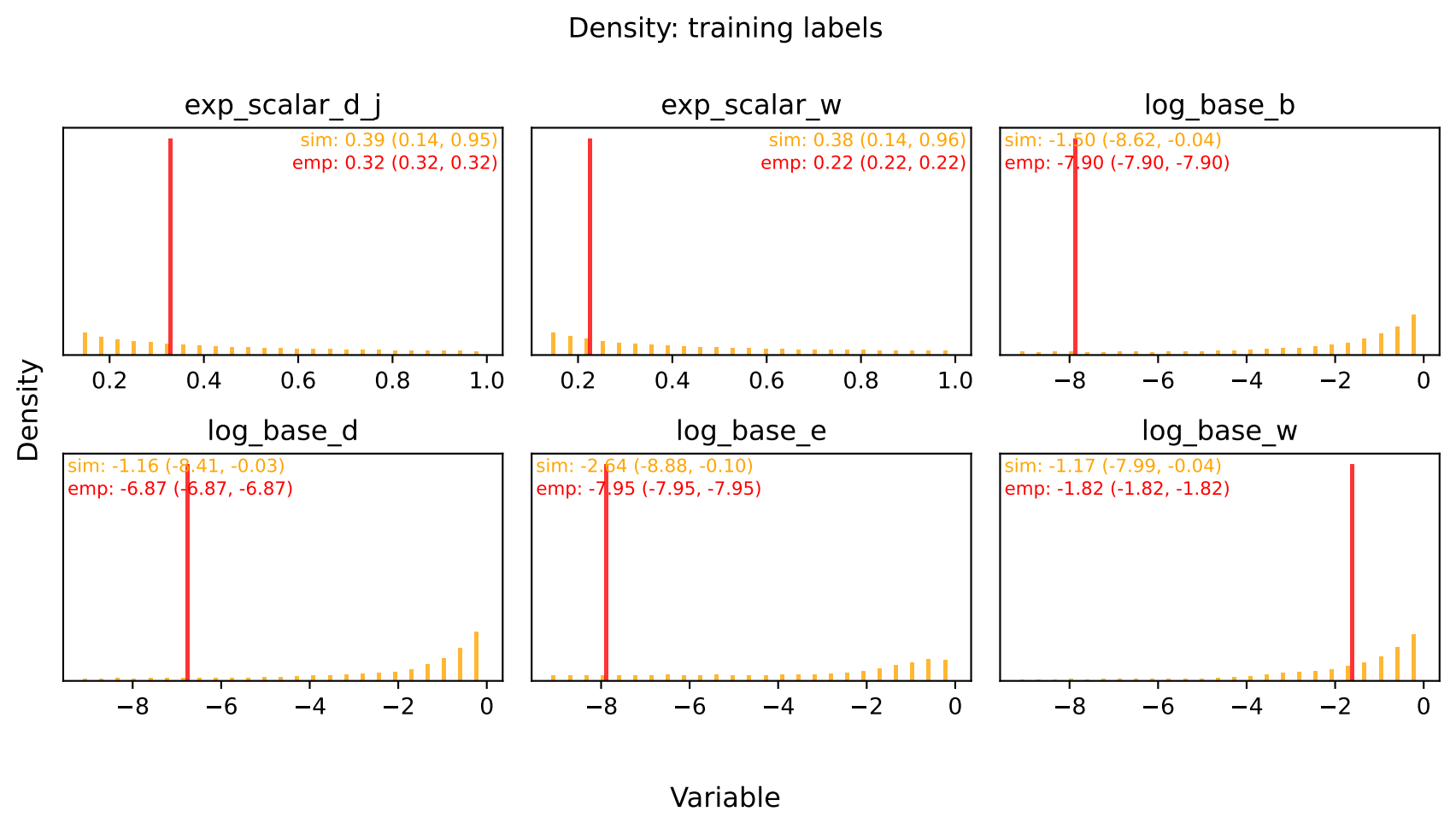

Estimate: sim.empirical.0
0
−2
−4
−6
−8
7.74E-01
3.22E-01
1.38E-01
-1.38E+00
-1.82E+00
-2.70E+00
-3.40E+00
-6.00E+00
-6.07E+00
-6.87E+00
-7.95E+00
-7.90E+00
-8.80E+00
-9.00E+00
-9.41E+00
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_w
exp_scalar_d_j

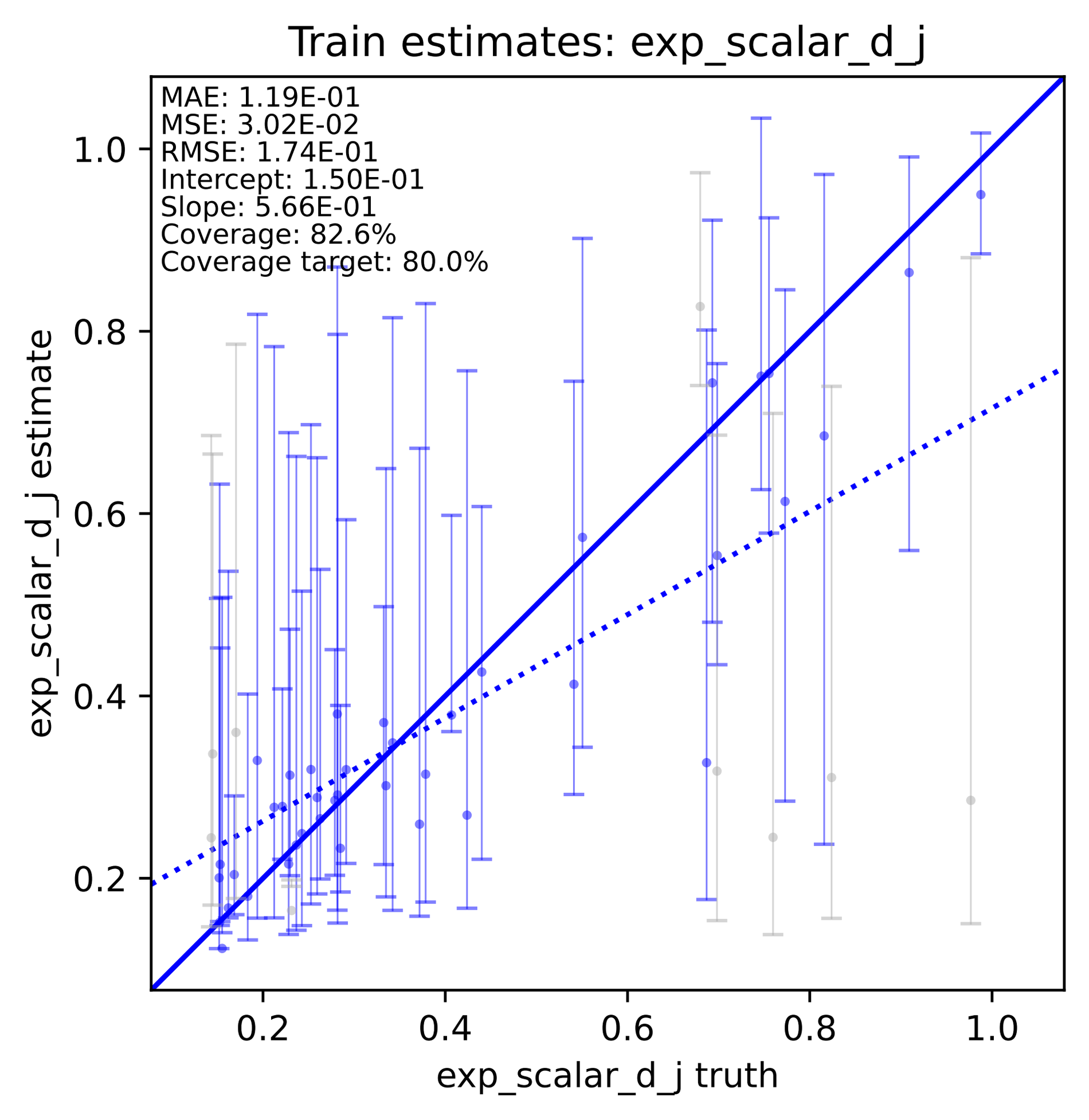

PCA: training aux. data
PC2 (27%)
% variance
PC
PC3 (18%)
PC4 (2%)
PC1 (47%)
PC2 (27%)
PC3 (18%)

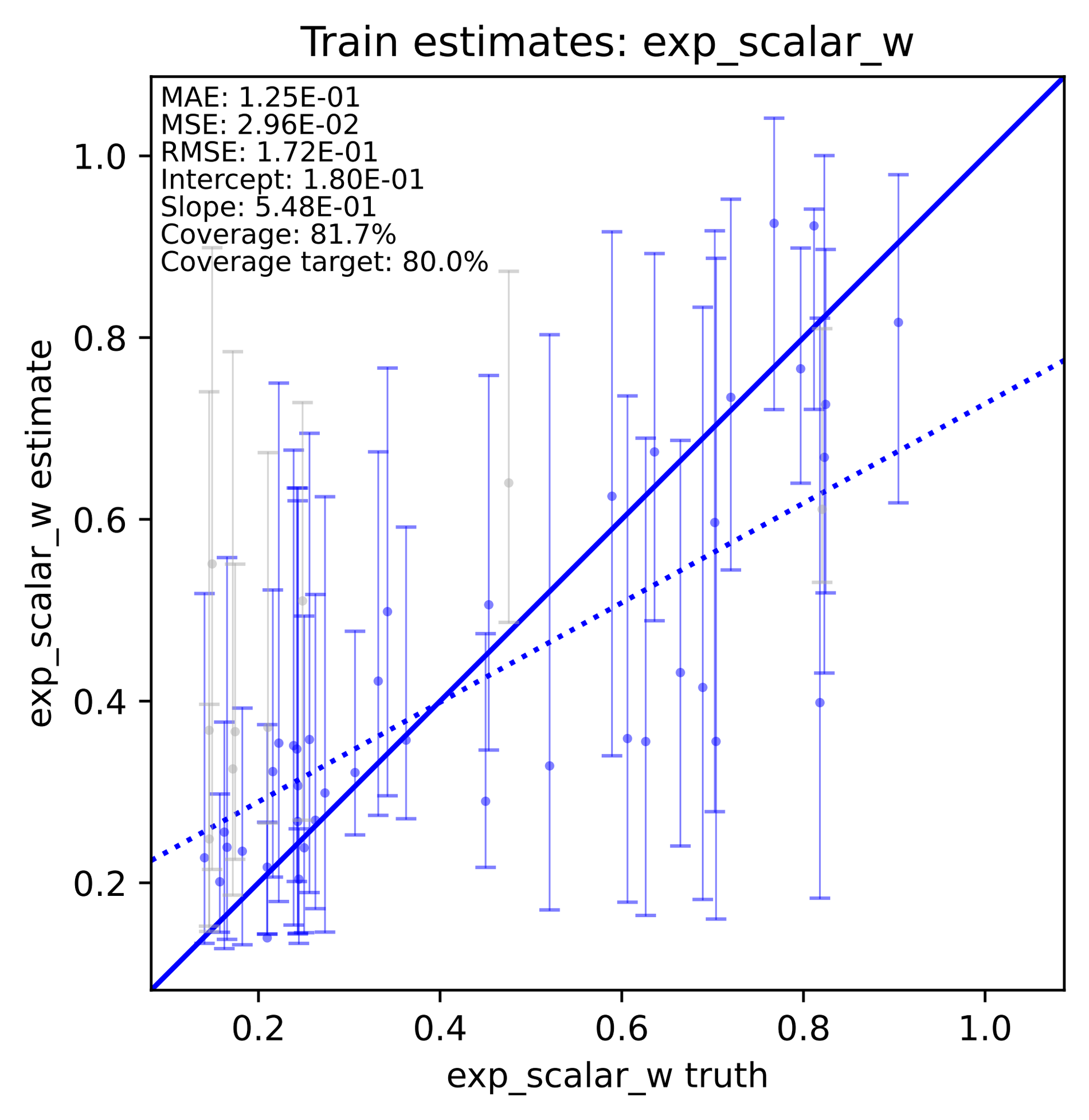

PCA: training labels
PC2 (17%)
% variance
PC
PC3 (15%)
PC4 (15%)
PC1 (34%)
PC2 (17%)
PC3 (15%)

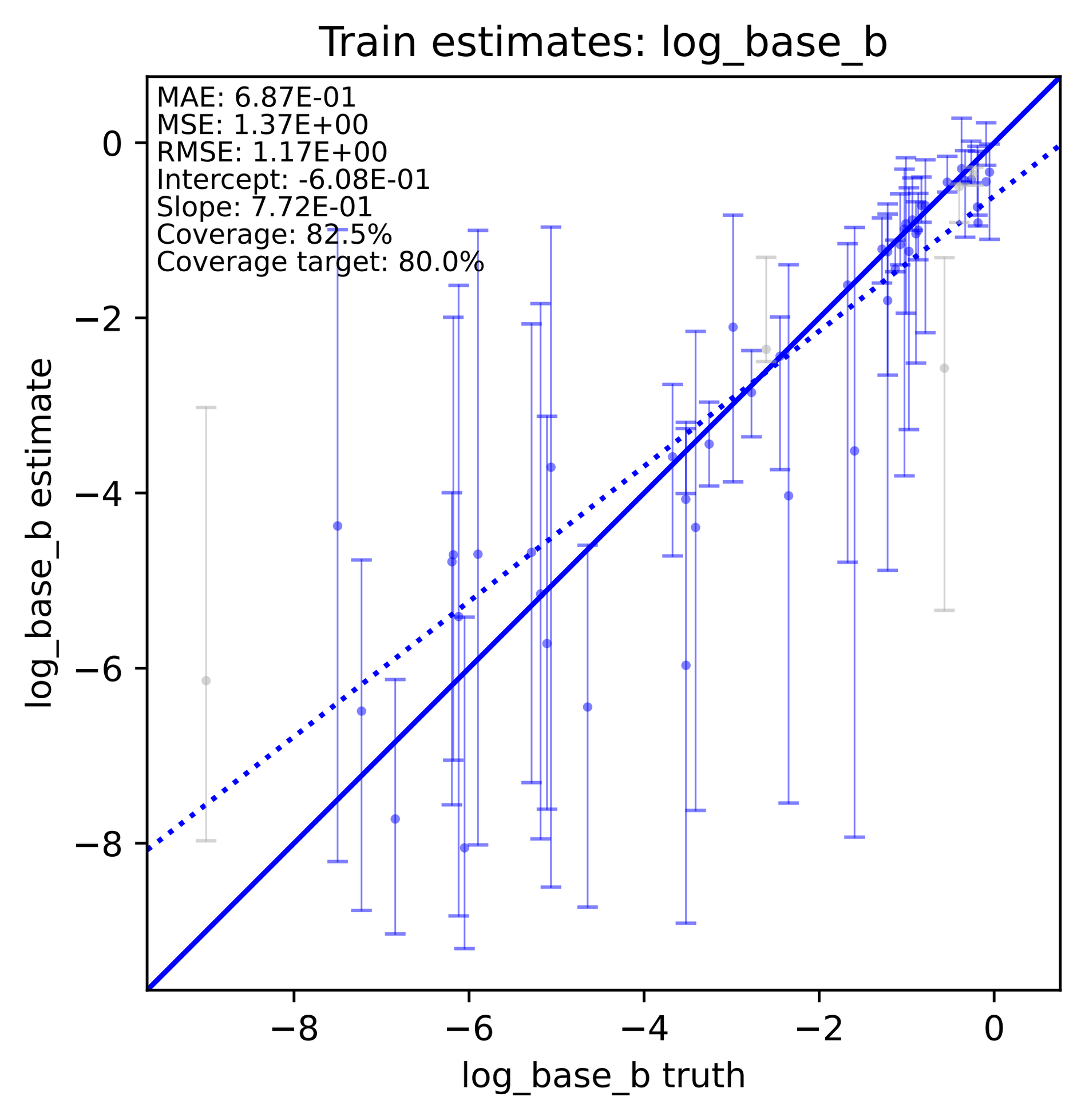

PCA: training phy. data
PC2 (9%)
% variance
PC
PC3 (2%)
PC4 (1%)
PC1 (25%)
PC2 (9%)
PC3 (2%)

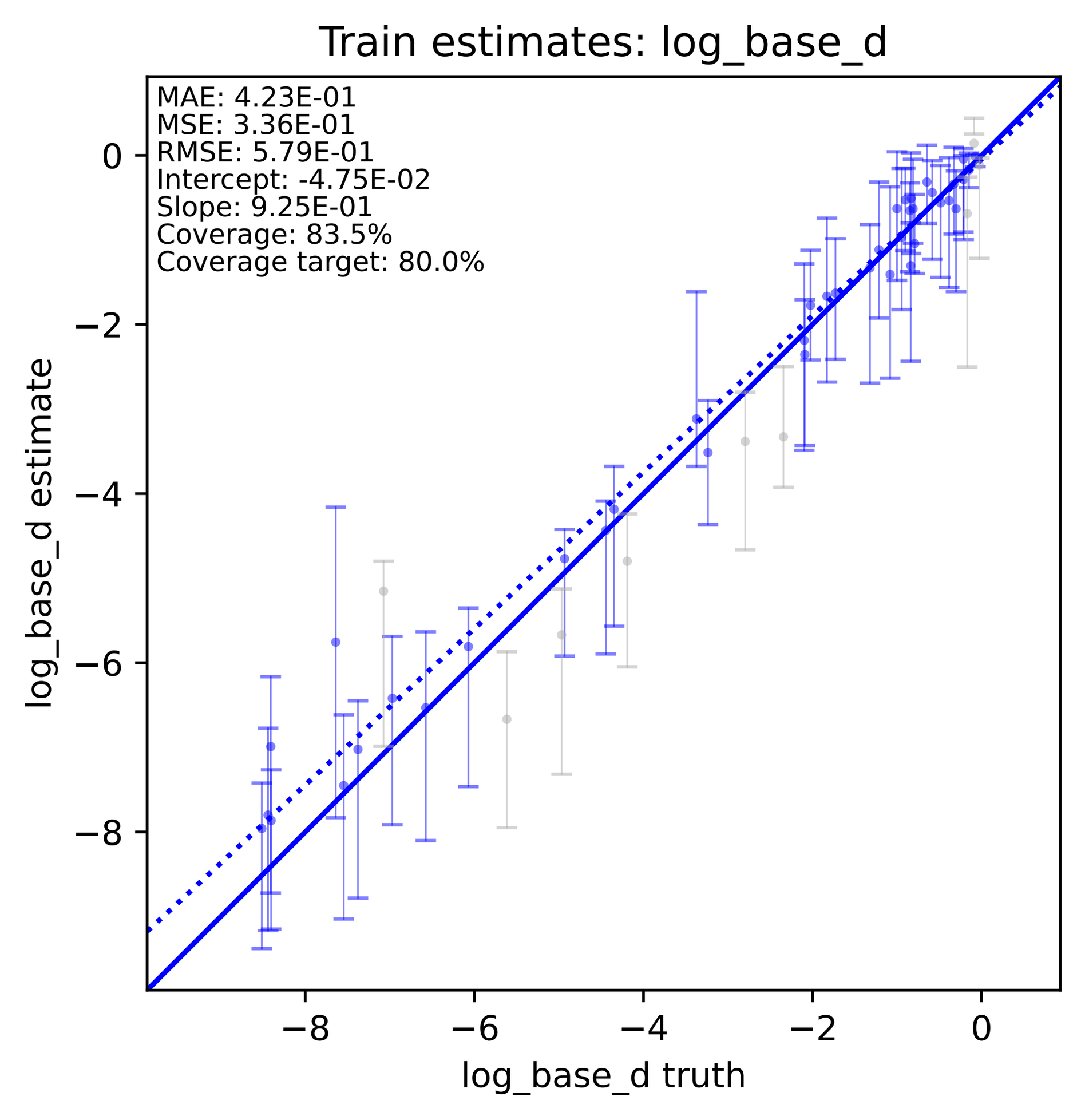

Density: training aux. data
N_bar
sim: 0.95 (0.60, 1.05)
emp: 0.91 (0.91, 0.91)
age_var
sim: 1.68 (-0.15, 2.86)
emp: 2.53 (2.53, 2.53)
brlen_var
sim: 1.19 (-0.72, 2.56)
emp: 2.45 (2.45, 2.45)
colless
sim: -1.32 (-1.63, -0.44)
emp: -0.57 (-0.57, -0.57)
f_dat_0
sim: 4.24 (0.92, 175.84)
emp: 3.95 (3.95, 3.95)
f_dat_1
sim: 2.30 (1.01, 44.40)
emp: 1.05 (1.05, 1.05)
log10_B1
sim: 1.93 (0.76, 2.20)
emp: 1.26 (1.26, 1.26)
log10_age_mean
sim: 0.81 (-0.04, 1.58)
emp: 1.52 (1.52, 1.52)
log10_brlen_mean
sim: 0.55 (-0.33, 1.33)
emp: 1.24 (1.24, 1.24)
log10_root_age
sim: 1.47 (0.60, 1.98)
emp: 1.88 (1.88, 1.88)
log10_tree_length
sim: 2.84 (1.95, 3.66)
emp: 3.11 (3.11, 3.11)
n_dat_0
sim: 6.70 (3.61, 9.01)
emp: 5.01 (5.01, 5.01)
n_dat_1
sim: 5.91 (3.04, 7.66)
emp: 3.69 (3.69, 3.69)
num_taxa
sim: 159.00 (11.00, 293.00)
emp: 38.00 (38.00, 38.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.29 (-0.45, -0.10)
emp: -0.83 (-0.83, -0.83)
Density
Variable

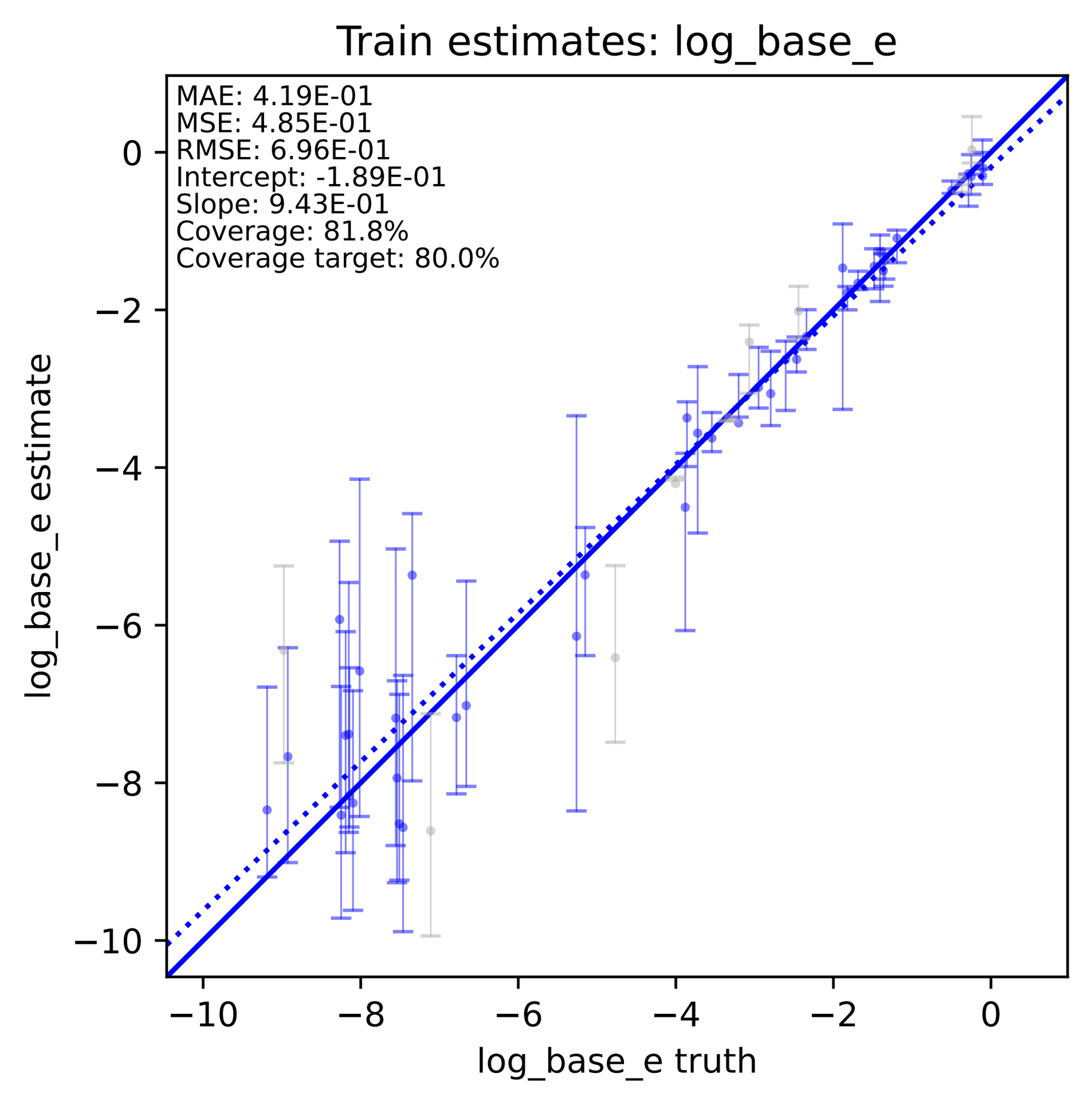

Density: training labels
exp_scalar_d_j
sim: 0.39 (0.14, 0.95)
emp: 0.32 (0.32, 0.32)
exp_scalar_w
sim: 0.38 (0.14, 0.96)
emp: 0.22 (0.22, 0.22)
log_base_b
sim: -1.50 (-8.62, -0.04)
emp: -7.90 (-7.90, -7.90)
log_base_d
sim: -1.16 (-8.41, -0.03)
emp: -6.87 (-6.87, -6.87)
log_base_e
sim: -2.64 (-8.88, -0.10)
emp: -7.95 (-7.95, -7.95)
log_base_w
sim: -1.17 (-7.99, -0.04)
emp: -1.82 (-1.82, -1.82)
Density
Variable

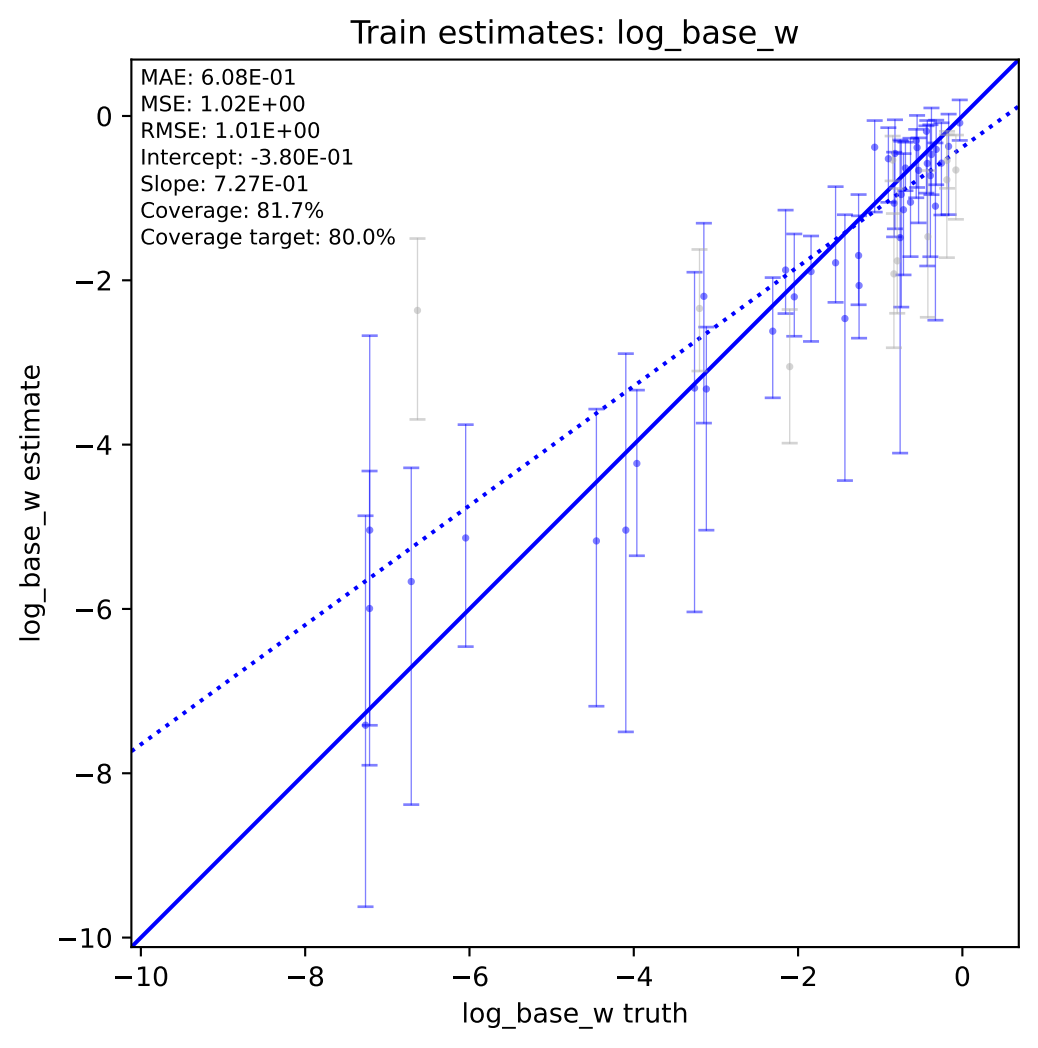

Train estimates: exp_scalar_d_j
MAE: 1.19E-01
MSE: 3.02E-02
RMSE: 1.74E-01
Intercept: 1.50E-01
Slope: 5.66E-01
Coverage: 82.6%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

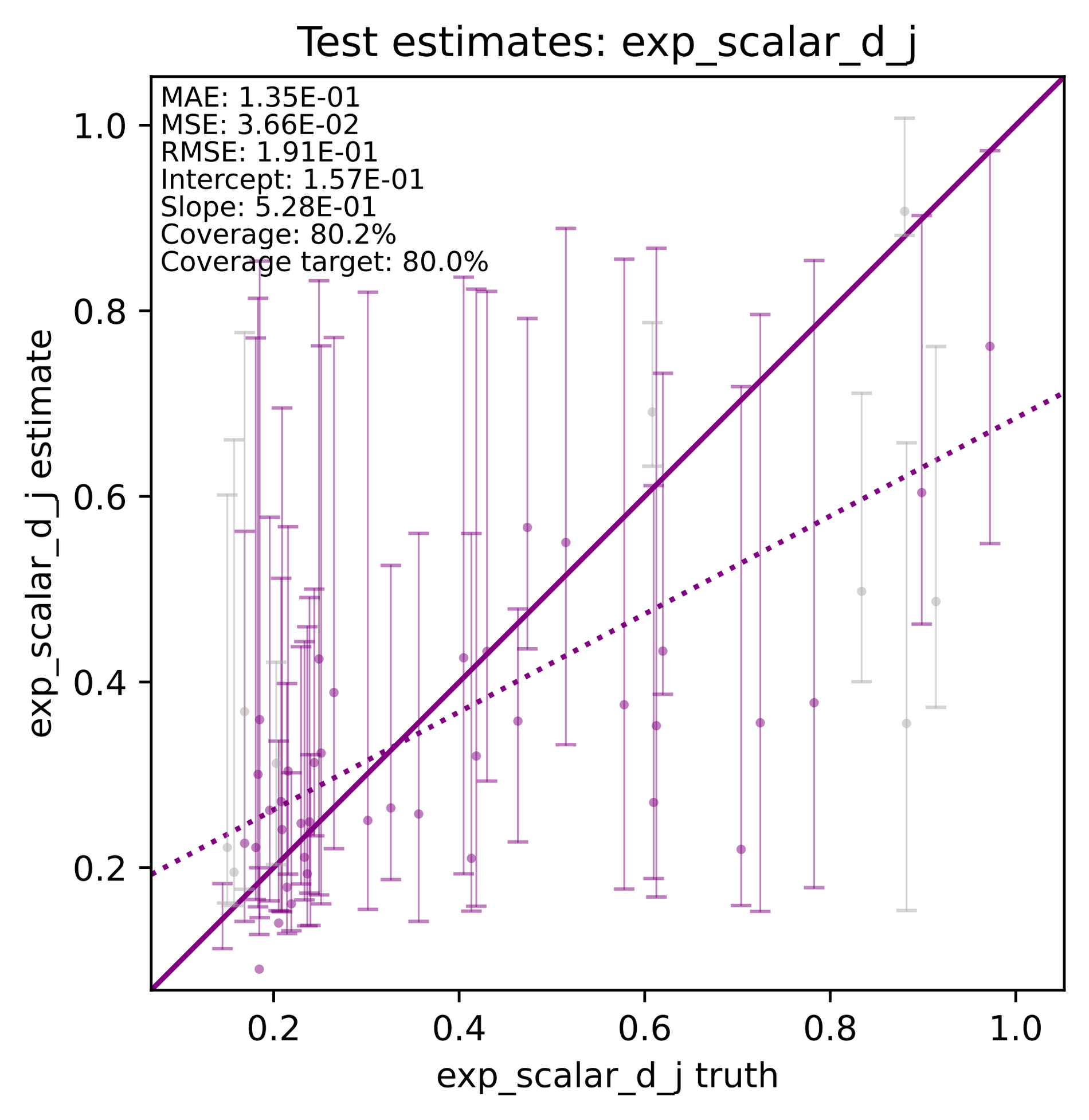

Train estimates: exp_scalar_w
MAE: 1.25E-01
MSE: 2.96E-02
RMSE: 1.72E-01
Intercept: 1.80E-01
Slope: 5.48E-01
Coverage: 81.7%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

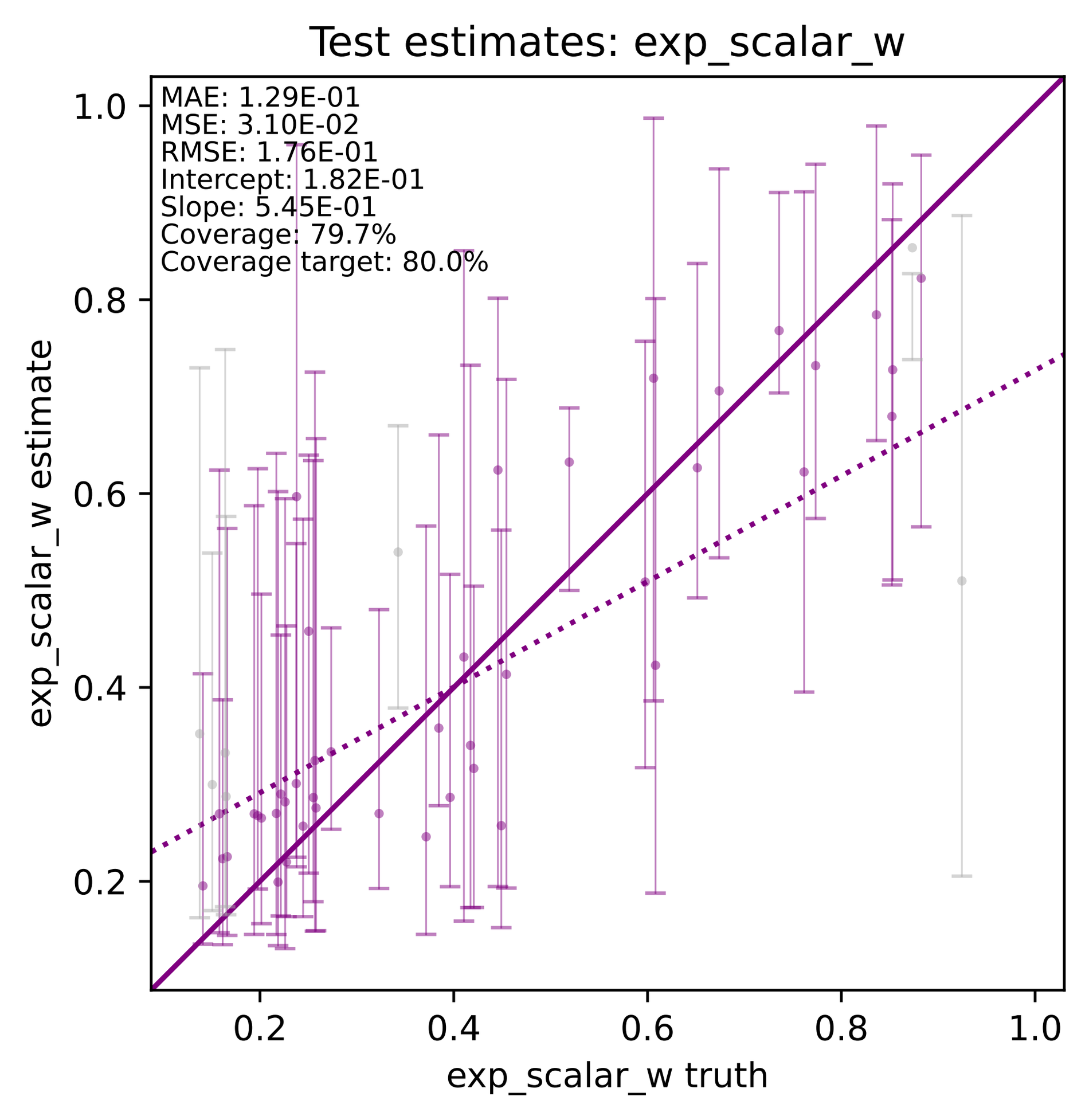

Train estimates: log_base_b
MAE: 6.87E-01
MSE: 1.37E+00
RMSE: 1.17E+00
Intercept: -6.08E-01
Slope: 7.72E-01
Coverage: 82.5%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

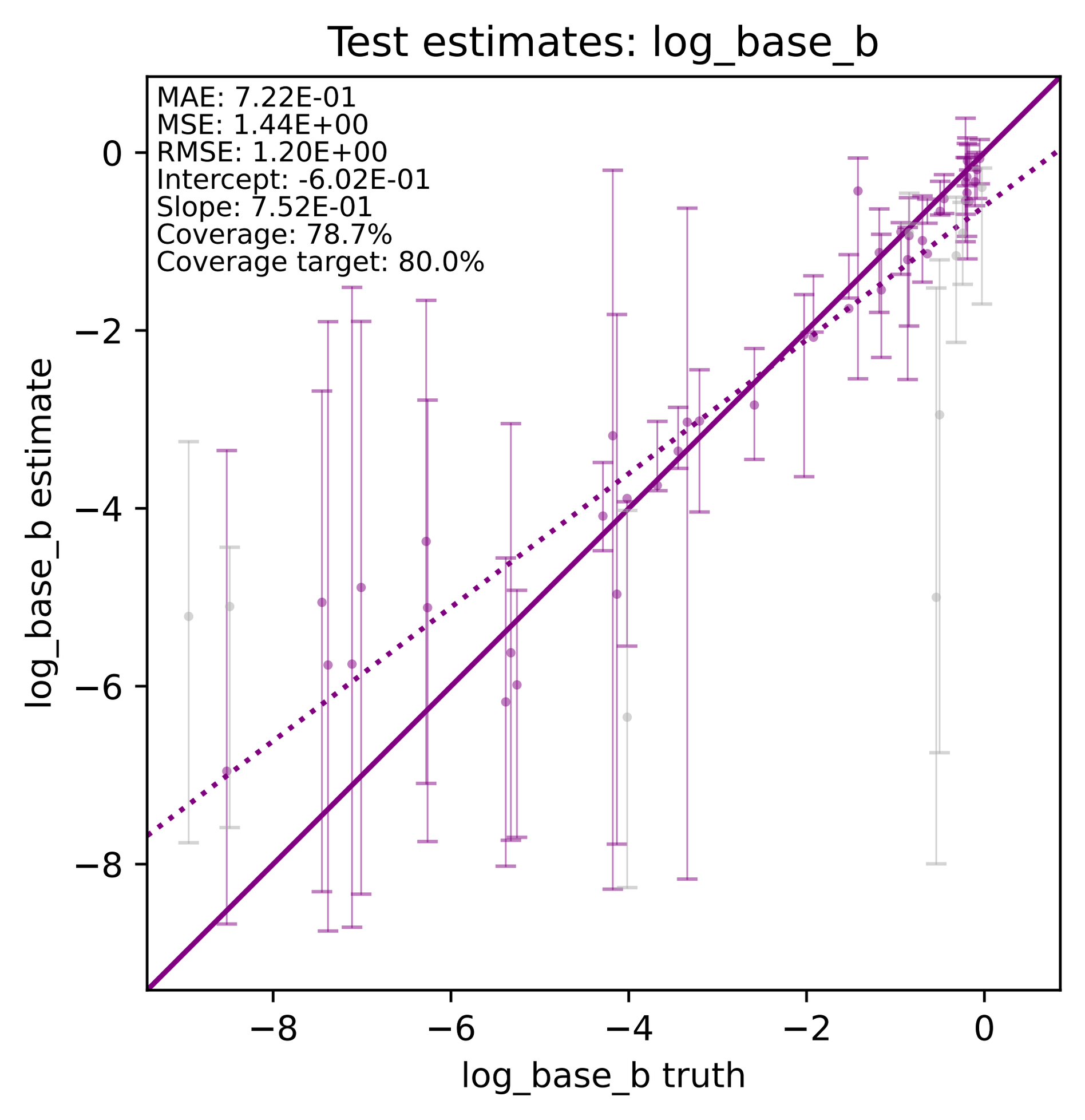

Train estimates: log_base_d
MAE: 4.23E-01
MSE: 3.36E-01
RMSE: 5.79E-01
Intercept: -4.75E-02
Slope: 9.25E-01
Coverage: 83.5%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

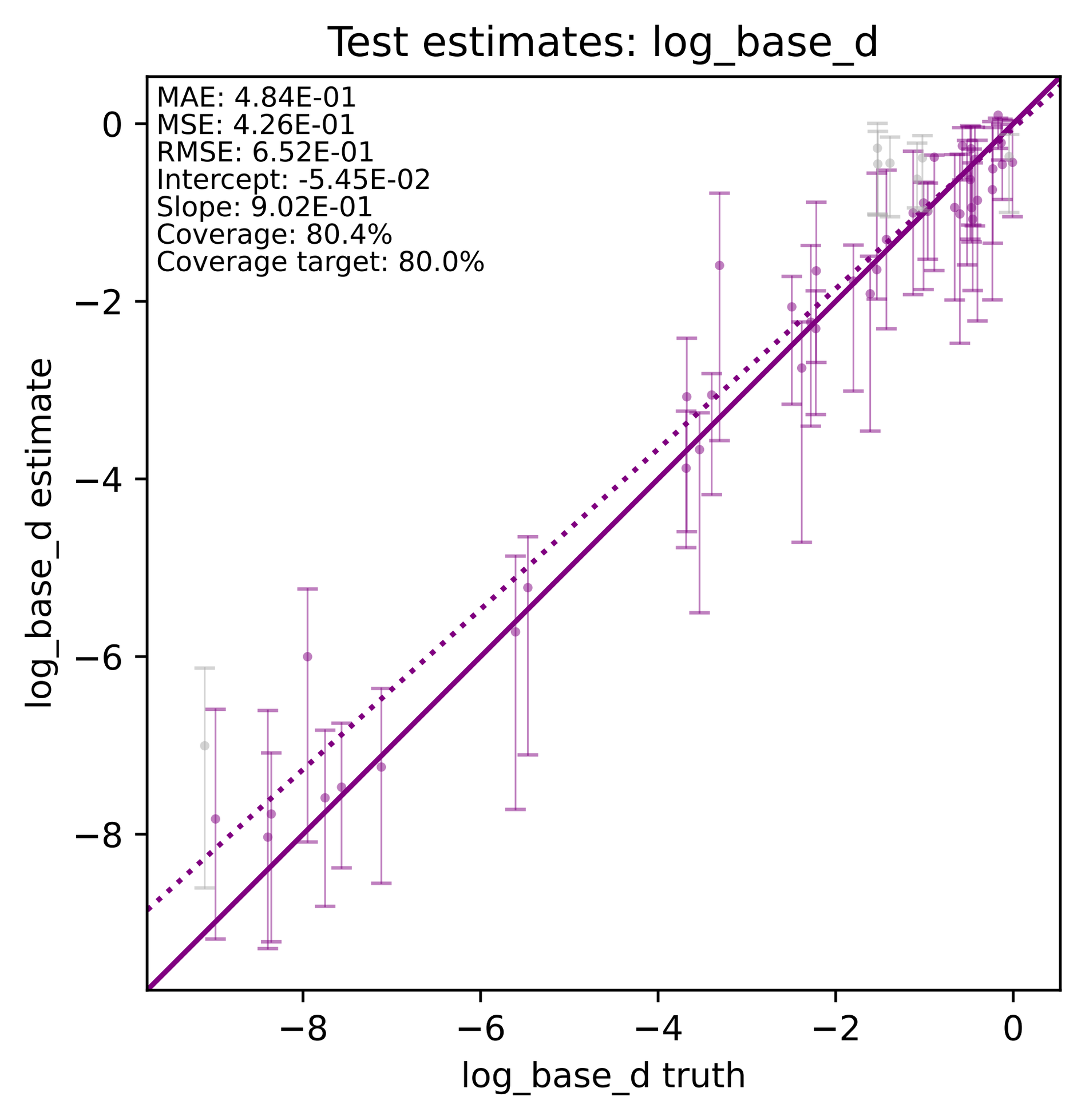

Train estimates: log_base_e
MAE: 4.19E-01
MSE: 4.85E-01
RMSE: 6.96E-01
Intercept: -1.89E-01
Slope: 9.43E-01
Coverage: 81.8%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

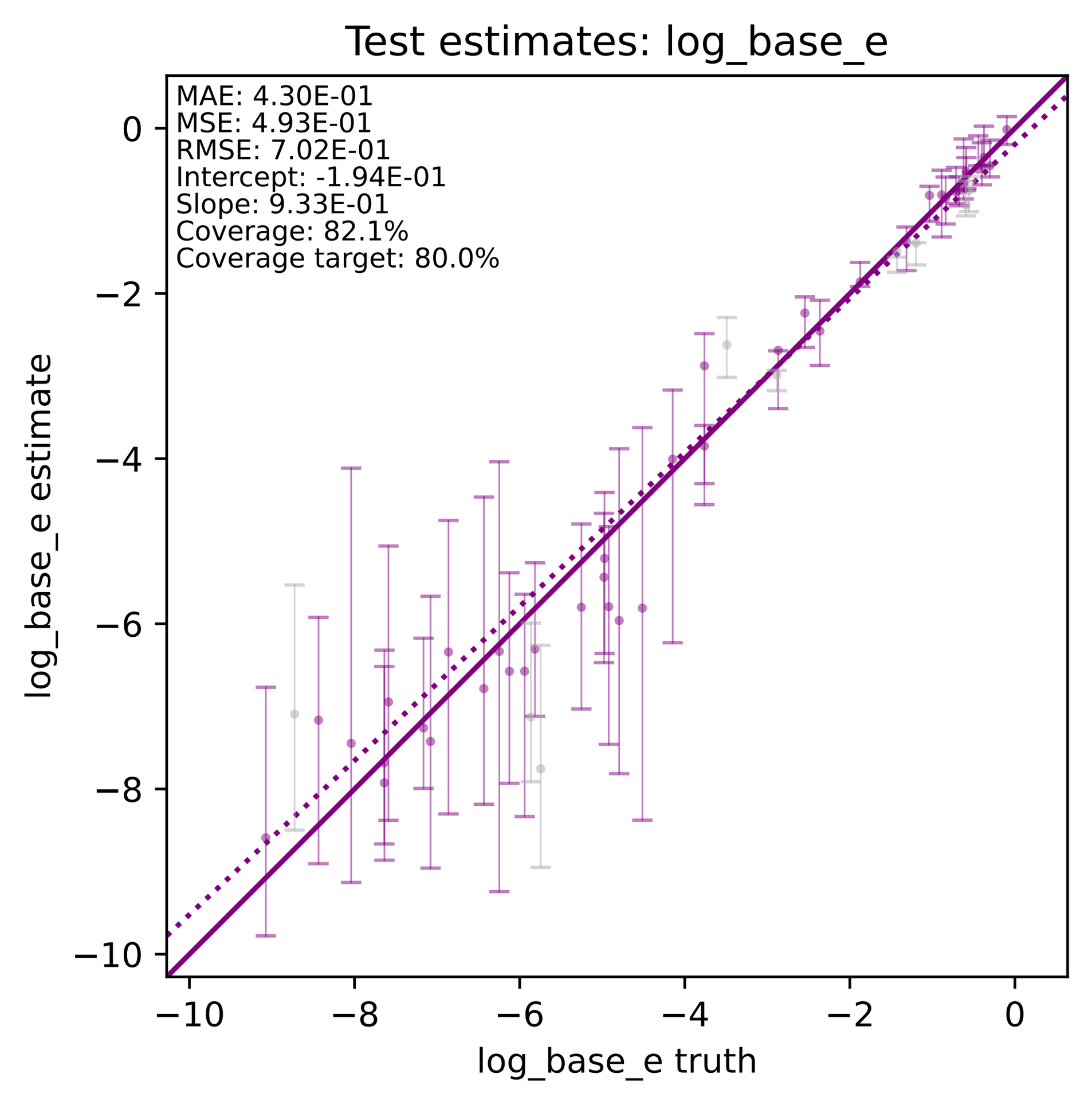

Train estimates: log_base_w
MAE: 6.08E-01
MSE: 1.02E+00
RMSE: 1.01E+00
Intercept: -3.80E-01
Slope: 7.27E-01
Coverage: 81.7%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

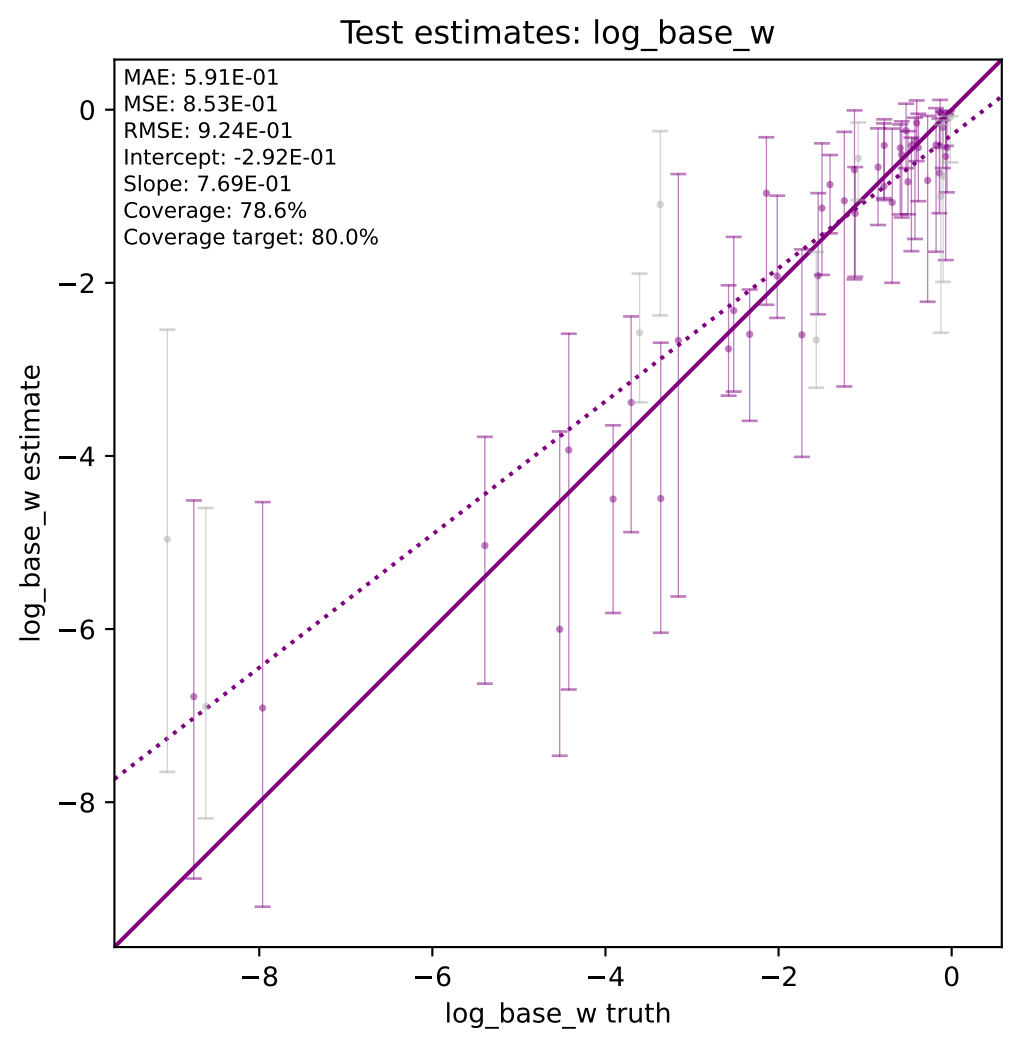

Test estimates: exp_scalar_d_j
MAE: 1.35E-01
MSE: 3.66E-02
RMSE: 1.91E-01
Intercept: 1.57E-01
Slope: 5.28E-01
Coverage: 80.2%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

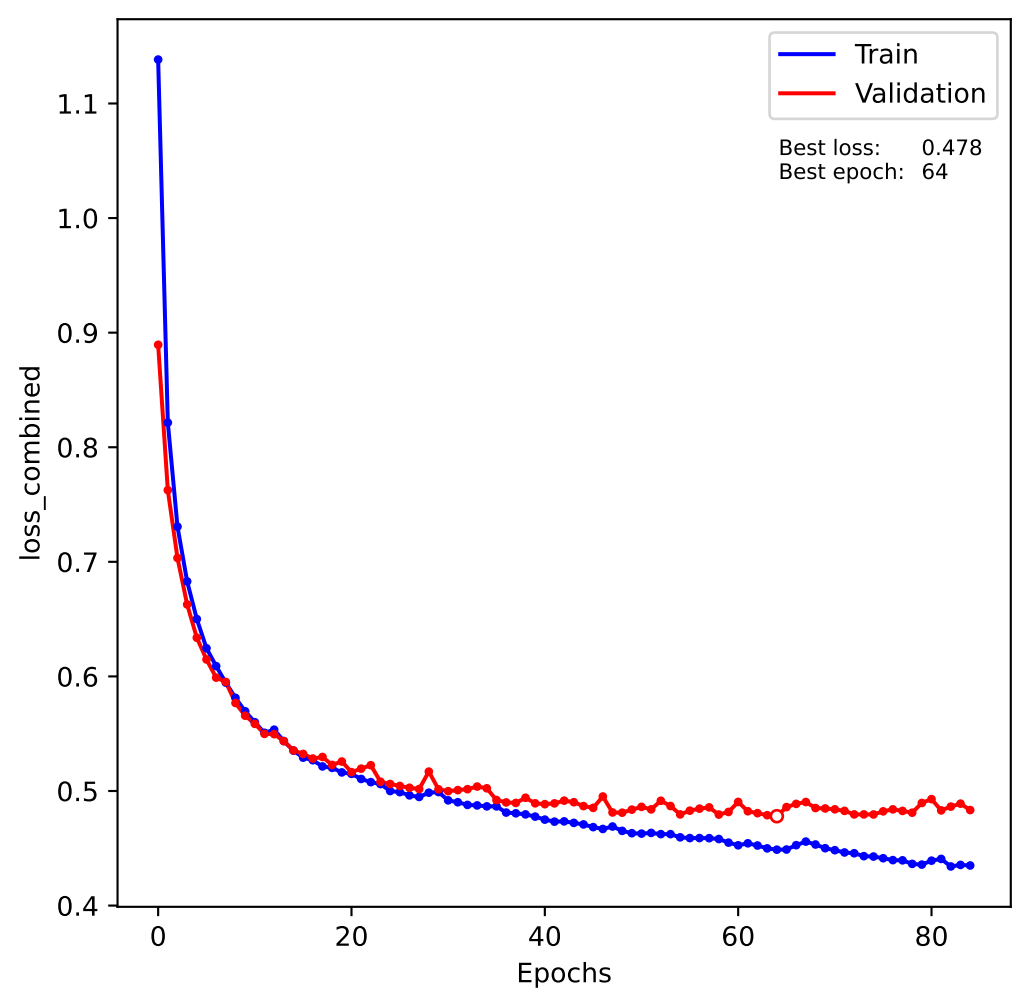

Test estimates: exp_scalar_w
MAE: 1.29E-01
MSE: 3.10E-02
RMSE: 1.76E-01
Intercept: 1.82E-01
Slope: 5.45E-01
Coverage: 79.7%
Coverage target: 80.0%
exp_scalar_w estimate
exp_scalar_w truth
0.2
0.4
0.6
0.8
1.0

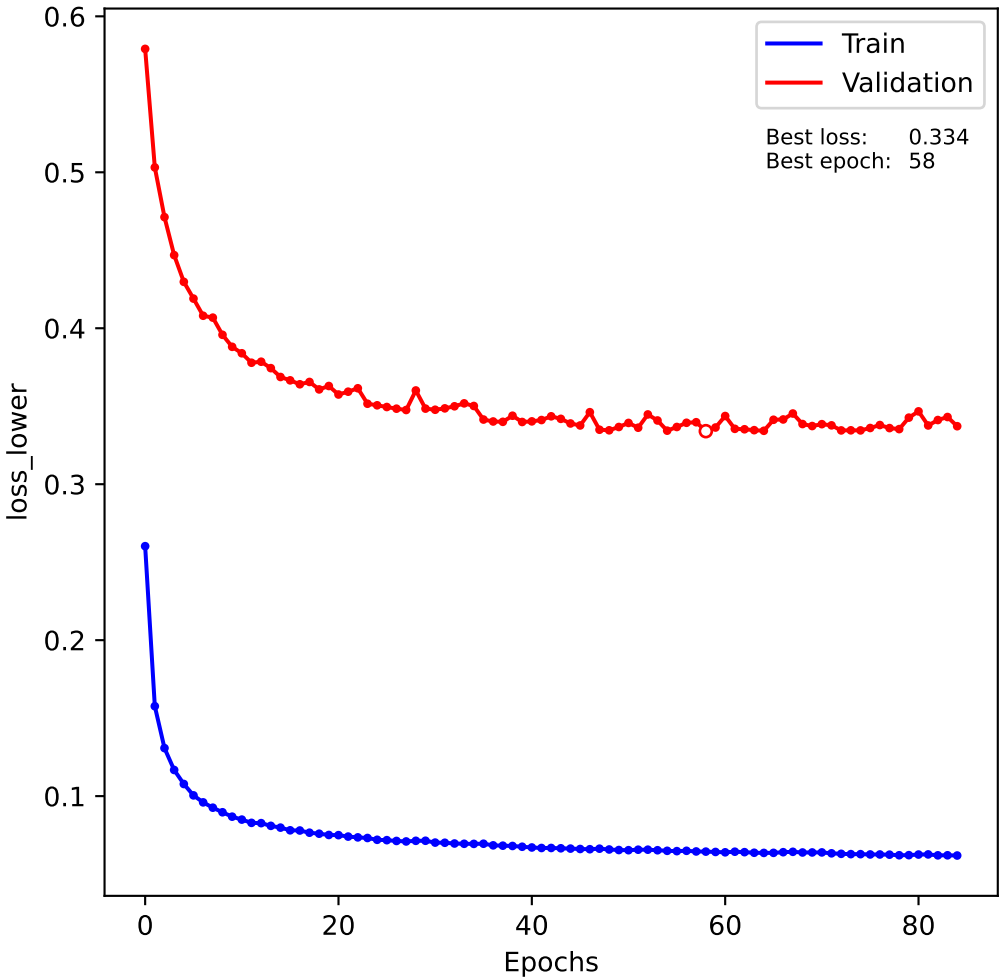

Test estimates: log_base_b
MAE: 7.22E-01
MSE: 1.44E+00
RMSE: 1.20E+00
Intercept: -6.02E-01
Slope: 7.52E-01
Coverage: 78.7%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

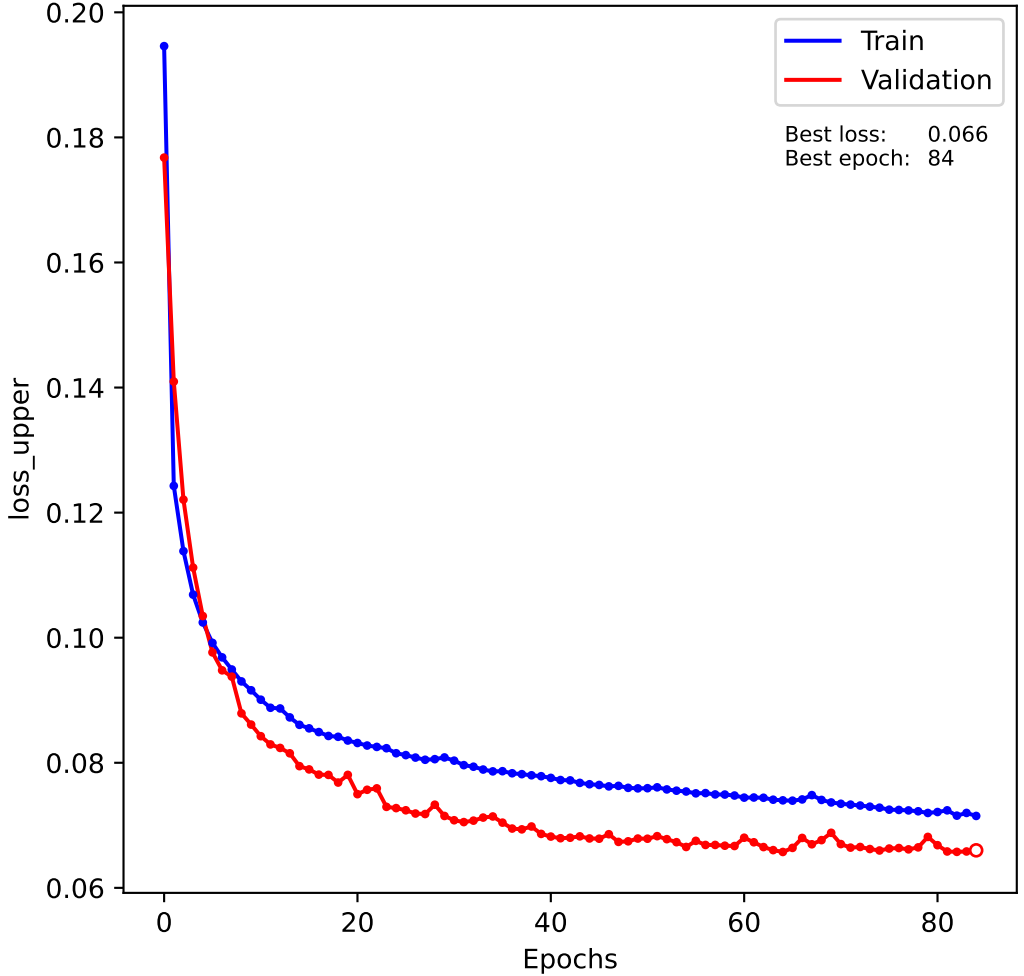

Test estimates: log_base_d
MAE: 4.84E-01
MSE: 4.26E-01
RMSE: 6.52E-01
Intercept: -5.45E-02
Slope: 9.02E-01
Coverage: 80.4%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

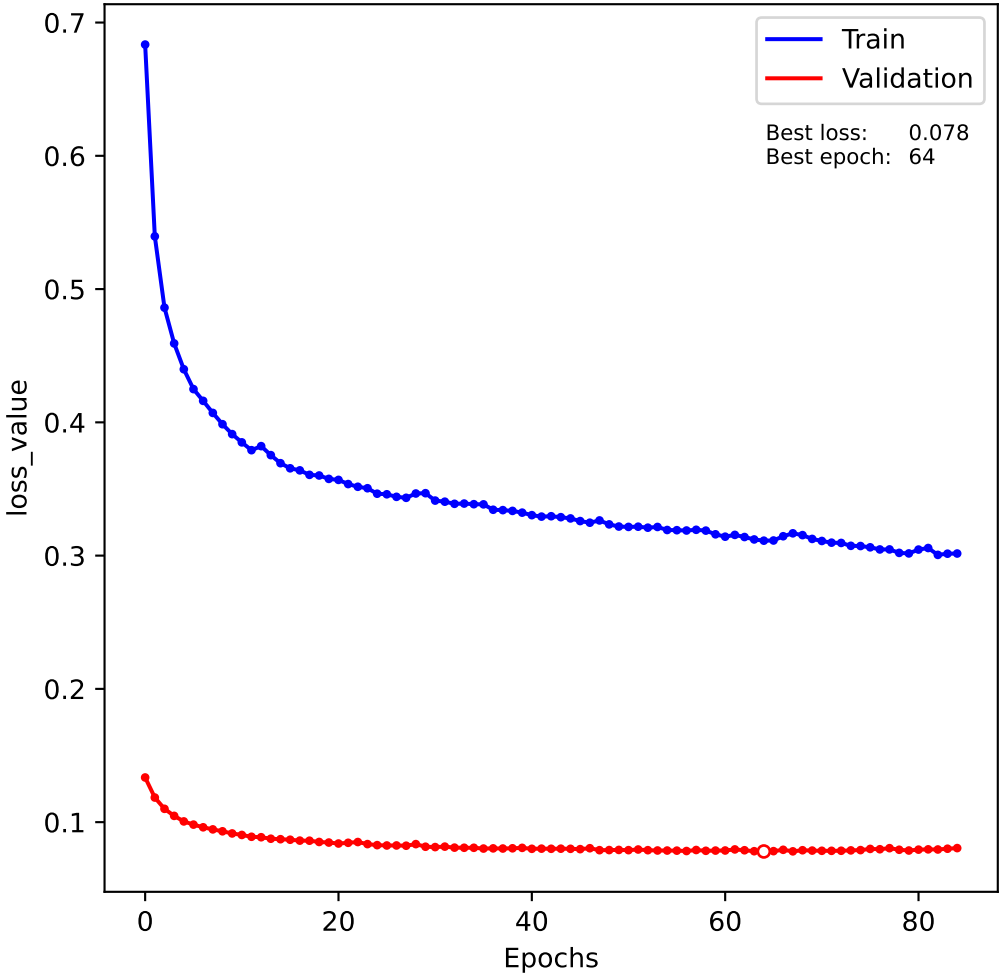

Test estimates: log_base_e
MAE: 4.30E-01
MSE: 4.93E-01
RMSE: 7.02E-01
Intercept: -1.94E-01
Slope: 9.33E-01
Coverage: 82.1%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

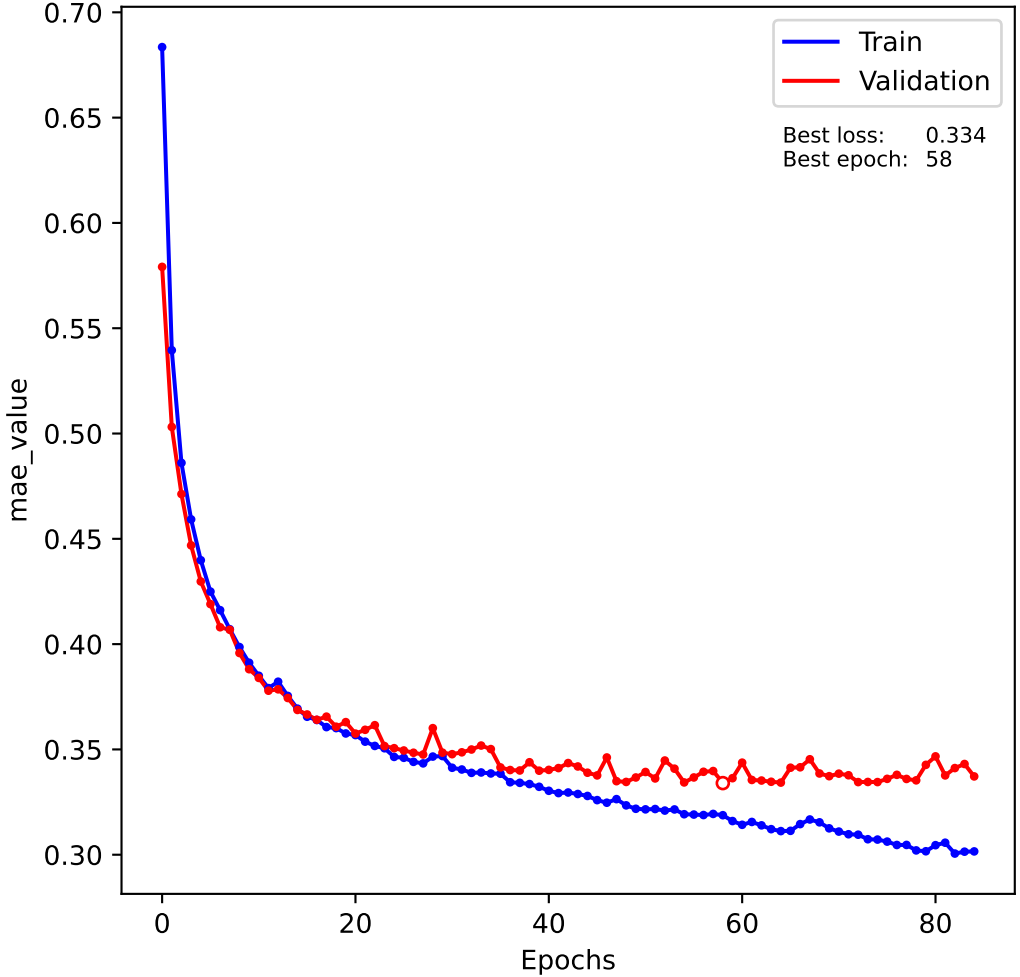

Test estimates: log_base_w
MAE: 5.91E-01
MSE: 8.53E-01
RMSE: 9.24E-01
Intercept: -2.92E-01
Slope: 7.69E-01
Coverage: 78.6%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8

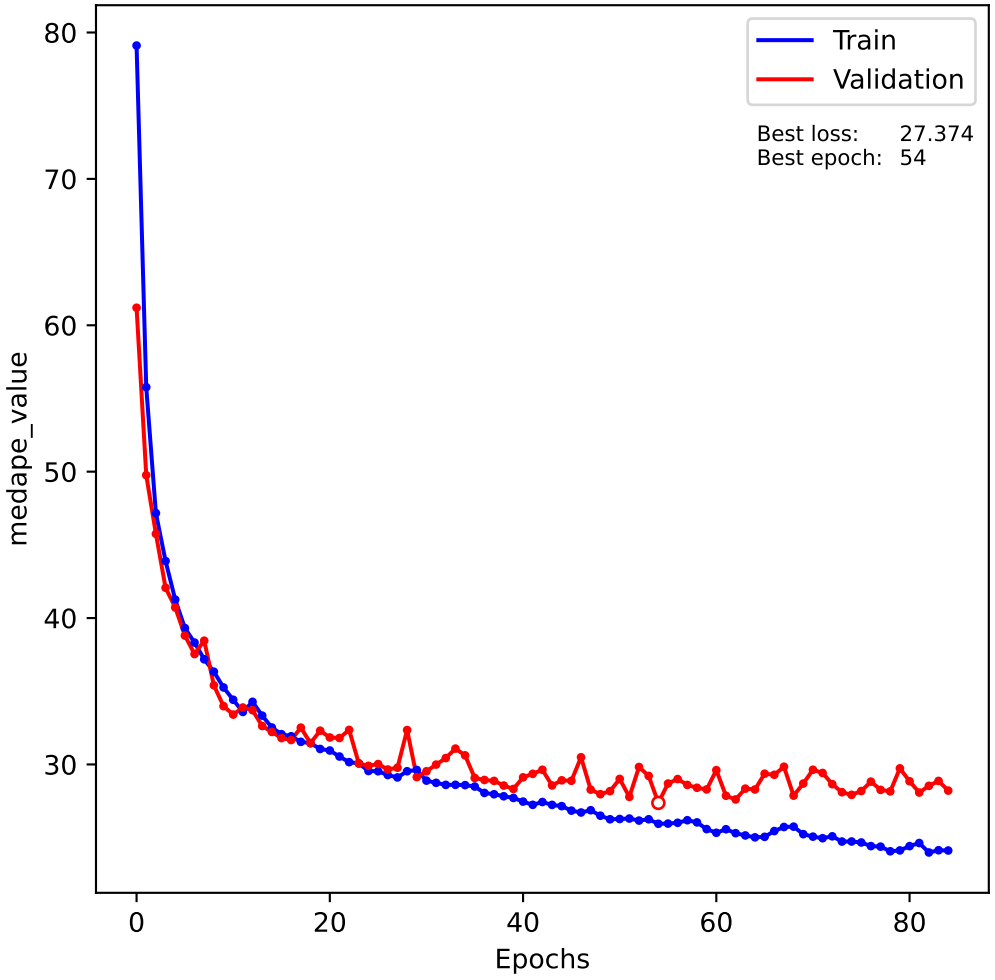

Train
Validation
Best loss: 0.478
Best epoch: 64
loss_combined
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0
20
40
60
80
Epochs

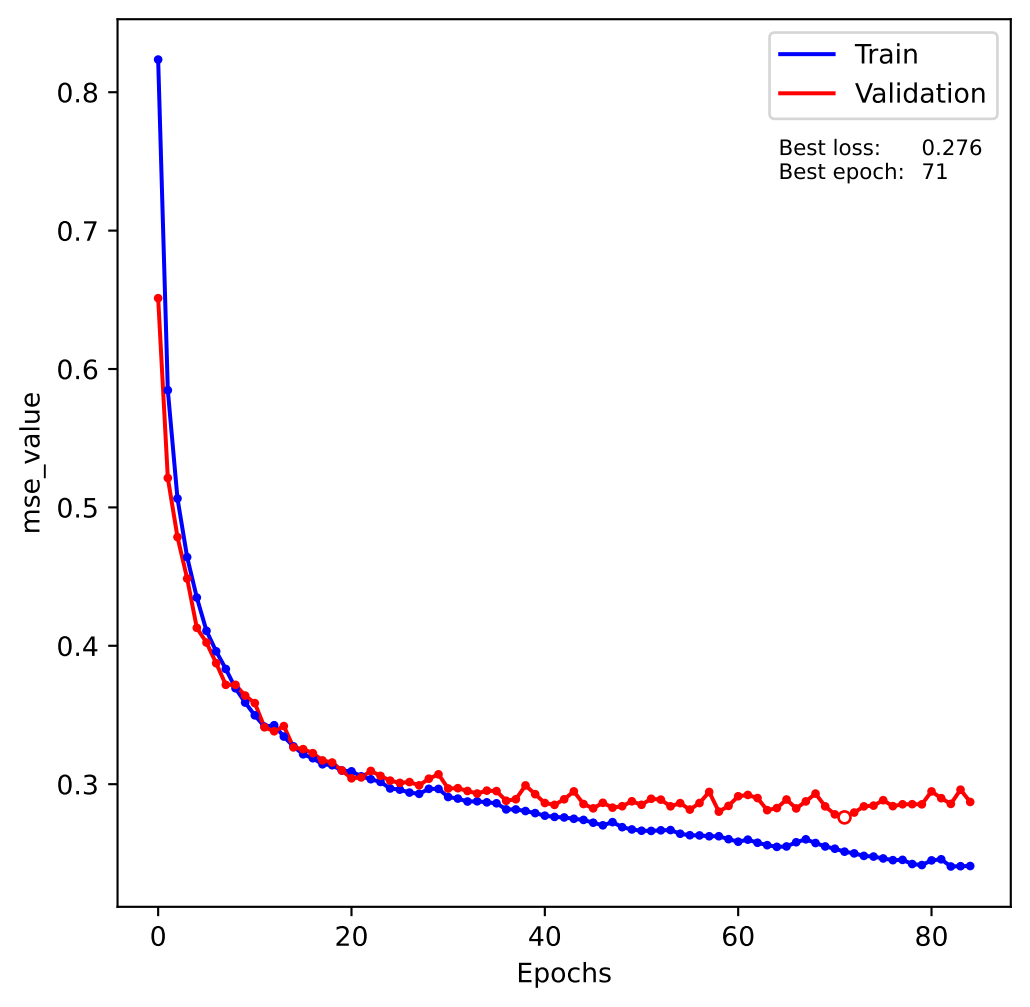

Train
Validation
Best loss: 0.334
Best epoch: 58
loss_lower
Epochs
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80

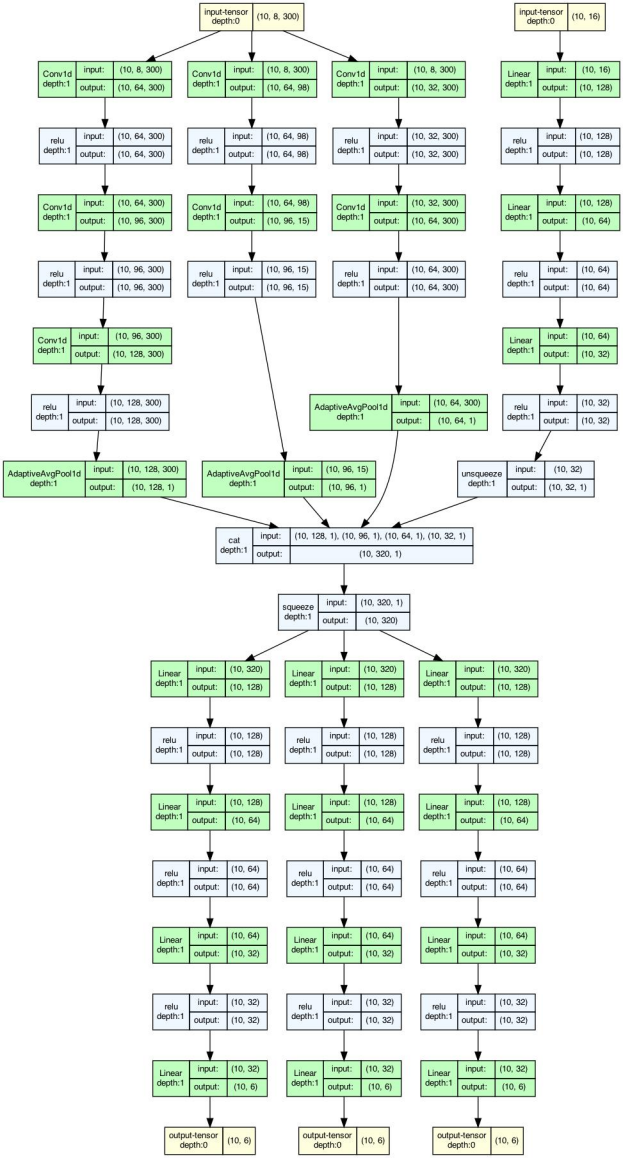

Train
Validation
Best loss: 0.066
Best epoch: 84
loss_upper
Epochs
0.20
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0
20
40
60
80


Train
Validation
Best loss: 0.078
Best epoch: 64
loss_value
Epochs
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
20
40
60
80


Train
Validation
Best loss: 0.334
Best epoch: 58
mae_value
Epochs
0.70
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0
20
40
60
80


Train
Validation
Best loss: 27.374
Best epoch: 54
medape_value
Epochs
80
70
60
50
40
30
0
20
40
60
80


Train
Validation
Best loss: 0.276
Best epoch: 71
mse_value
0.8
0.7
0.6
0.5
0.4
0.3
0
20
40
60
80
Epochs


input-tensor depth:0 (10, 8, 300)
input-tensor depth:0 (10, 16)
Conv1d depth:1 input: (10, 8, 300) output: (10, 64, 300)
Conv1d depth:1 input: (10, 8, 300) output: (10, 64, 98)
Conv1d depth:1 input: (10, 8, 300) output: (10, 32, 300)
Linear depth:1 input: (10, 16) output: (10, 128)
relu depth:1 input: (10, 64, 300) output: (10, 64, 300)
relu depth:1 input: (10, 64, 98) output: (10, 64, 98)
relu depth:1 input: (10, 32, 300) output: (10, 32, 300)
relu depth:1 input: (10, 128) output: (10, 128)
Conv1d depth:1 input: (10, 64, 300) output: (10, 96, 300)
Conv1d depth:1 input: (10, 64, 98) output: (10, 96, 15)
Conv1d depth:1 input: (10, 32, 300) output: (10, 64, 300)
Linear depth:1 input: (10, 128) output: (10, 64)
relu depth:1 input: (10, 96, 300) output: (10, 96, 300)
relu depth:1 input: (10, 96, 15) output: (10, 96, 15)
relu depth:1 input: (10, 64, 300) output: (10, 64, 300)
relu depth:1 input: (10, 64) output: (10, 64)
Conv1d depth:1 input: (10, 96, 300) output: (10, 128, 300)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 128, 300) output: (10, 128, 300)
AdaptiveAvgPool1d depth:1 input: (10, 64, 300) output: (10, 64, 1)
relu depth:1 input: (10, 32) output: (10, 32)
AdaptiveAvgPool1d depth:1 input: (10, 128, 300) output: (10, 128, 1)
AdaptiveAvgPool1d depth:1 input: (10, 96, 15) output: (10, 96, 1)
unsqueeze depth:1 input: (10, 32) output: (10, 32, 1)
cat depth:1 input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1) output: (10, 320, 1)
squeeze depth:1 input: (10, 320, 1) output: (10, 320)
Linear depth:1 input: (10, 320) output: (10, 128)
Linear depth:1 input: (10, 320) output: (10, 128)
Linear depth:1 input: (10, 320) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
relu depth:1 input: (10, 128) output: (10, 128)
Linear depth:1 input: (10, 128) output: (10, 64)
Linear depth:1 input: (10, 128) output: (10, 64)
Linear depth:1 input: (10, 128) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
relu depth:1 input: (10, 64) output: (10, 64)
Linear depth:1 input: (10, 64) output: (10, 32)
Linear depth:1 input: (10, 64) output: (10, 32)
Linear depth:1 input: (10, 64) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
relu depth:1 input: (10, 32) output: (10, 32)
Linear depth:1 input: (10, 32) output: (10, 6)
Linear depth:1 input: (10, 32) output: (10, 6)
Linear depth:1 input: (10, 32) output: (10, 6)
output-tensor depth:0 (10, 6)
output-tensor depth:0 (10, 6)
output-tensor depth:0 (10, 6)